

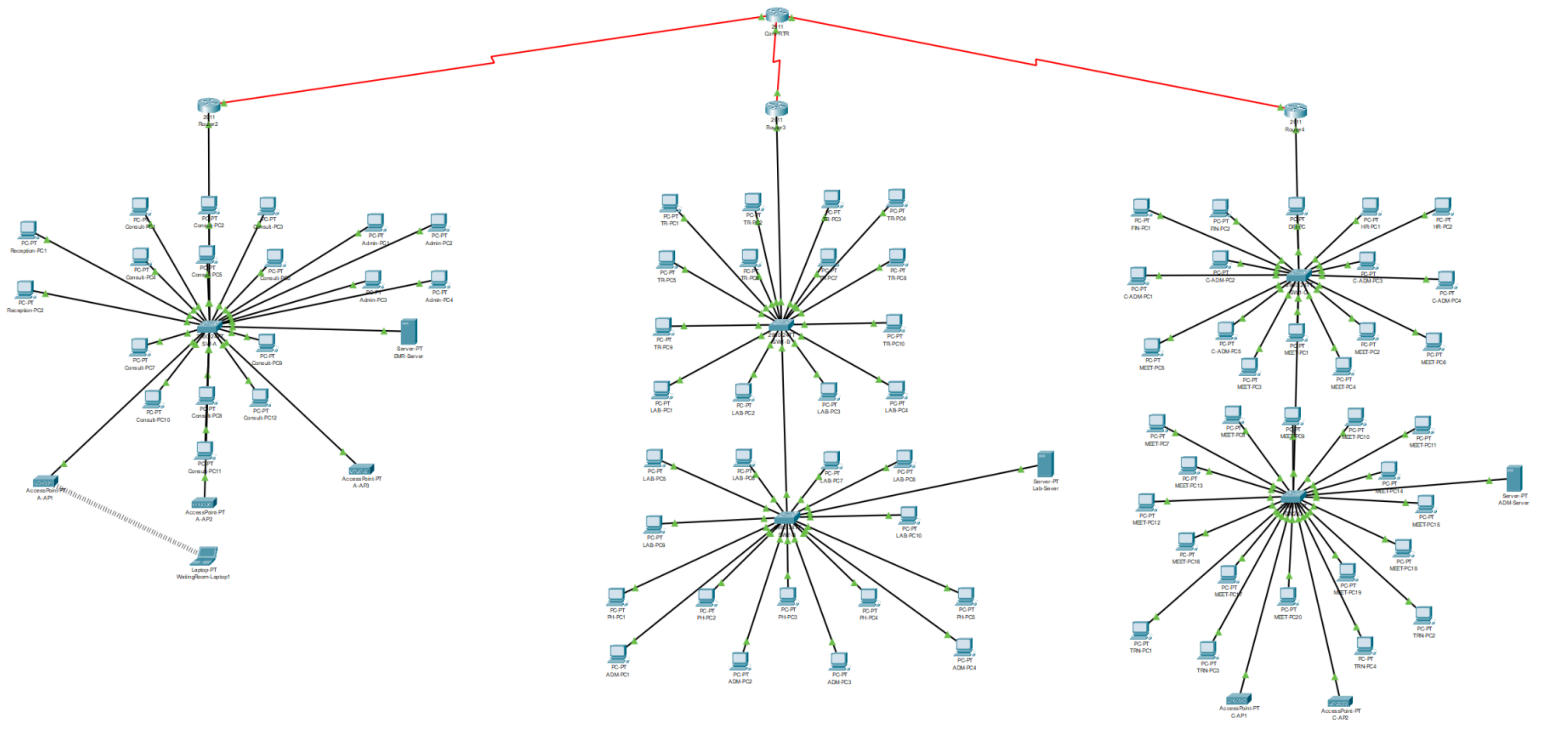


Network Design for a Hospital

Group 10

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◆ Final Design of the Hospital Network



❖ Introduction:

This report presents the design and implementation of a complete network infrastructure for a new hospital consisting of three buildings:

Building A: Outpatient clinics and reception.

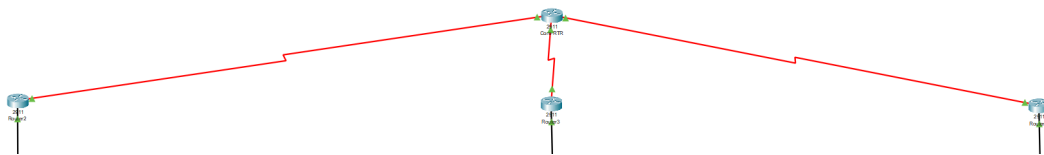
Building B: Treatment rooms, labs, and pharmacy.

Building C: Administration, finance, HR, and meeting areas.

The entire network was designed and simulated using Cisco Packet Tracer to verify connectivity and functionality before real deployment.

Each building contains appropriate devices based on its functional needs:

- **Building A:** Consultation rooms, reception, admin offices, EMR server, waiting area Wi-Fi.
- **Building B:** Treatment rooms, laboratories with dedicated lab PCs, pharmacy PCs, and a lab server.
- **Building C:** Administrative offices, finance, HR, conference room with 20 PCs, training rooms, café Wi-Fi, and an admin server.



All three buildings are connected using a *central Core Router*. Each building has its own router, and all routers connect to the core via serial WAN links, enabling full communication between buildings.

The inter-building connections were implemented using serial WAN links (DCE/DTE) on HWIC-2T modules. Each link uses a /30 network for efficient point-to-point routing.

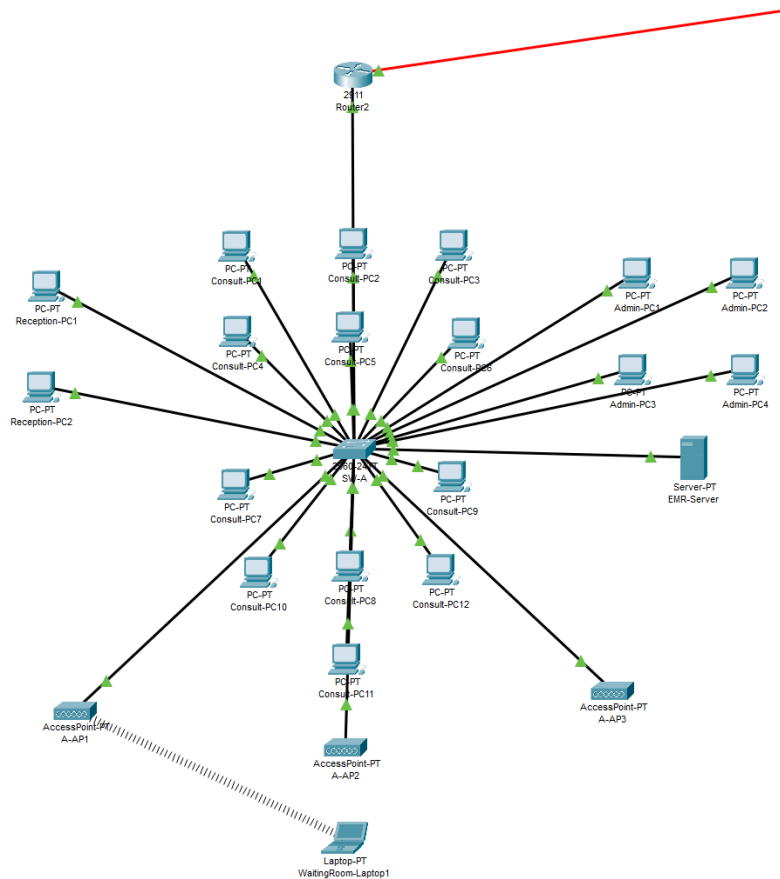
Each building was assigned a separate subnet to simplify management and routing:

- Building A → **10.10.10.0**
- Building B → **10.10.20.0**
- Building C → **10.10.30.0**

This prevents conflicts and ensures clear separation of departments.

Every building includes its own *2911 router* and one or more *2960 switches*. This allows internal LAN traffic to stay local while still supporting communication with other buildings.

- **Building A: Outpatient Clinics**

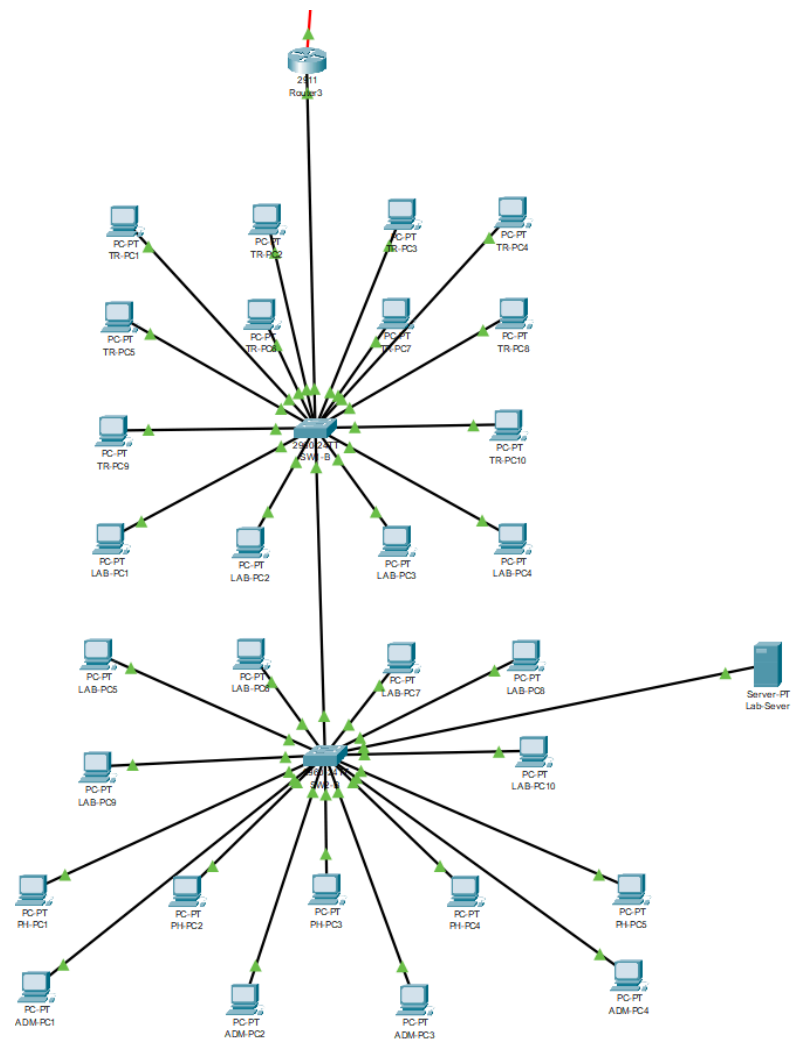


Requirements: 2 reception PCs, 6 waiting rooms with Wi-Fi, 12 consultation rooms, 4 admin offices, EMR support.

Devices implemented:

- 1 Router
- 1 Switch (24/48 ports)
- 2 Reception PCs
- 12 Consultation PCs
- 4 Admin PCs
- 1 EMR Server
- 2 Access Points for waiting room coverage
- 1 Laptop for Wi-Fi testing

- **Building B: Specialist Services and Laboratories, and pharmacy.**

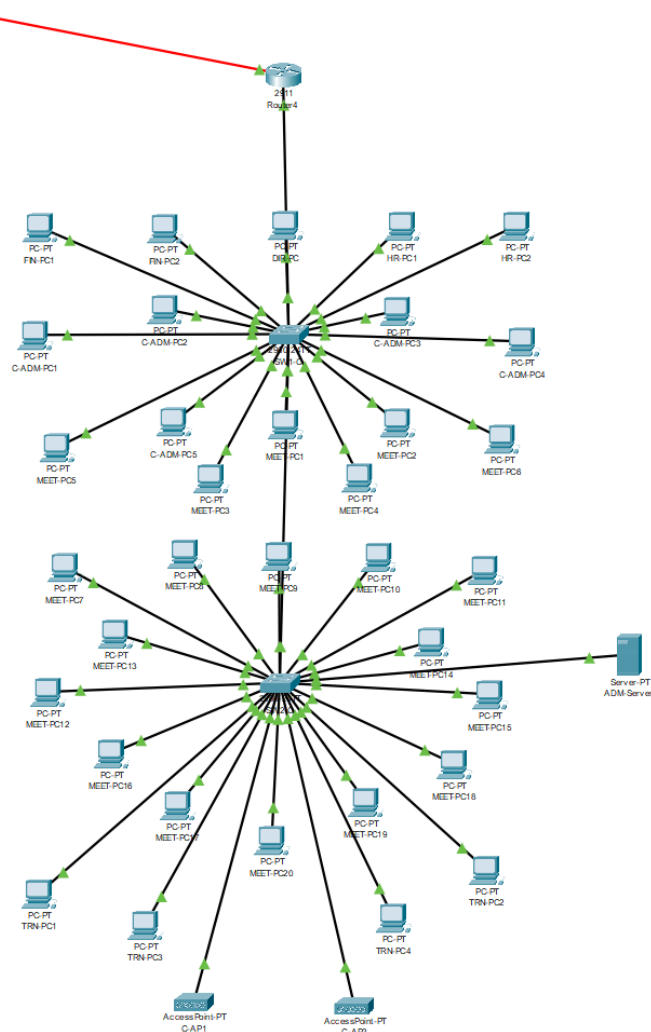


Requirements: 10 treatment rooms, 2 labs with 10 PCs, secure lab traffic, 5 pharmacy PCs, 4 admin PCs.

Devices implemented:

- 1 Router
- 2 Switches
- 10 Treatment Room PCs
- 10 Lab PCs
- 5 Pharmacy PCs
- 4 Admin PCs
- 1 Lab Server
- 1 Access Point

- **Building C: Administration and Support**



Requirements: Director office, Finance x2, HR x2, 5 admin offices, conference room (20 PCs), training rooms (2×2 PCs), staff café Wi-Fi.

Devices implemented:

- 1 Router
- 2 Switches
- 20 Conference PCs
- 4 Training PCs
- Director PC
- 2 Finance PCs
- 2 HR PCs
- 5 General admin PCs
- 1 Access Point for café
- 1 Admin Server

❖ Network Servers

- Building A : EMR_Server

Electronic Medical Records (EMR) system for doctors, reception, and admin staff.

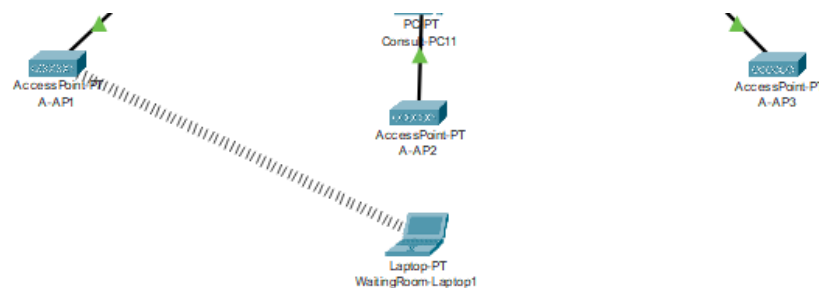
- Building B : Lab_Server

Laboratory Information System (LIS) and storage of lab test results.

- Building C : C-ADM-SRV

Administrative, HR, and finance systems and shared internal documents.

❖ Building A Access Points



Several wireless access points (AP1, AP2, AP3) were installed to provide Wi-Fi coverage in different areas.

There is a laptop connected wirelessly to AP1.

Each access point is configured with an SSID, a password, and DHCP so devices can get an IP address automatically when they connect.

❖ Building C Access Points



two wireless access points (C-AP1 and C-AP2) in Building C.

These access points were installed to provide Wi-Fi coverage in different areas of the building, such as the café. Each access point is configured with an SSID, a password, and DHCP so devices can automatically receive an IP address when they connect.

❖ IP Addressing Plan

Building	Network	Subnet Mask	Gateway
A	10.10.10.0	255.255.255.0	10.10.10.1
B	10.10.20.0	255.255.255.0	10.10.20.1
C	10.10.30.0	255.255.255.0	10.10.30.1

❖ Routing Configuration

Core Router

```
Router(config-if)#interface Serial0/3/0
Router(config-if)# ip address 192.168.10.1 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# no shutdown
```

Explanation:

- **Serial0/3/0** is the interface connecting to **Building A**
- *192.168.10.1* is the IP address of the **core side**
- *.1* for core, *.2* for Building A
- *clock rate 64000* is required on the **DCE** side (the clock-providing router)
- *no shutdown* activates the interface

```
Router(config-if)#interface Serial0/2/1
Router(config-if)# ip address 192.168.10.9 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# no shutdown
```

Explanation:

- This interface connects the Core Router to **Building C**
- IP *192.168.10.9* belongs to a different WAN subnet
- The *.9* is the core router, *.10* will be the building C router
- Clock rate again because Core Router provides timing


```

Router(config-if)#interface Serial0/3/1
Router(config-if)# ip address 192.168.10.5 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# no shutdown

```

Explanation:

- *Serial0/3/1* connects to **Building B**
- WAN network is *192.168.10.4*
- .5 is the core router's IP, .6 will be Building B's IP
- Clock rate because core is DCE

In addition to configuring the serial interfaces, static routing was used to enable inter-building communication. On the core router, a static route was added for each building subnet.

```

Router>show ip interface brief

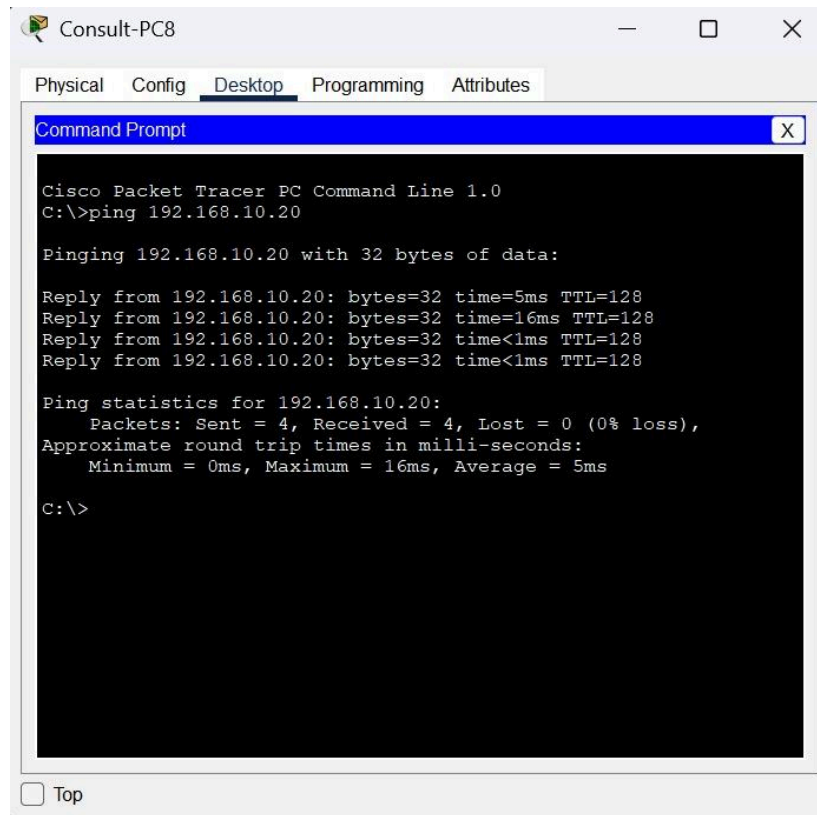
```

Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	unassigned	YES	unset	
administratively down	down			
GigabitEthernet0/1	unassigned	YES	unset	
administratively down	down			
GigabitEthernet0/2	unassigned	YES	unset	
administratively down	down			
Serial0/2/0	192.168.10.9	YES	manual	
administratively down	down			
Serial0/2/1	192.168.10.9	YES	manual	up
up				
Serial0/3/0	192.168.10.1	YES	manual	up
up				
Serial0/3/1	192.168.10.5	YES	manual	up
up				
Vlan1	unassigned	YES	unset	
administratively down	down			

As shown in the screenshot, the three serial interfaces (S0/3/0, S0/3/1, S0/2/1) appear as up/up. The up/up status indicates that the serial interface is physically active and logically operational, allowing traffic to pass between buildings. This confirms successful connectivity from the Core Router to Building A, Building B, and Building C.

❖ Ping tests

- Building A



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.20

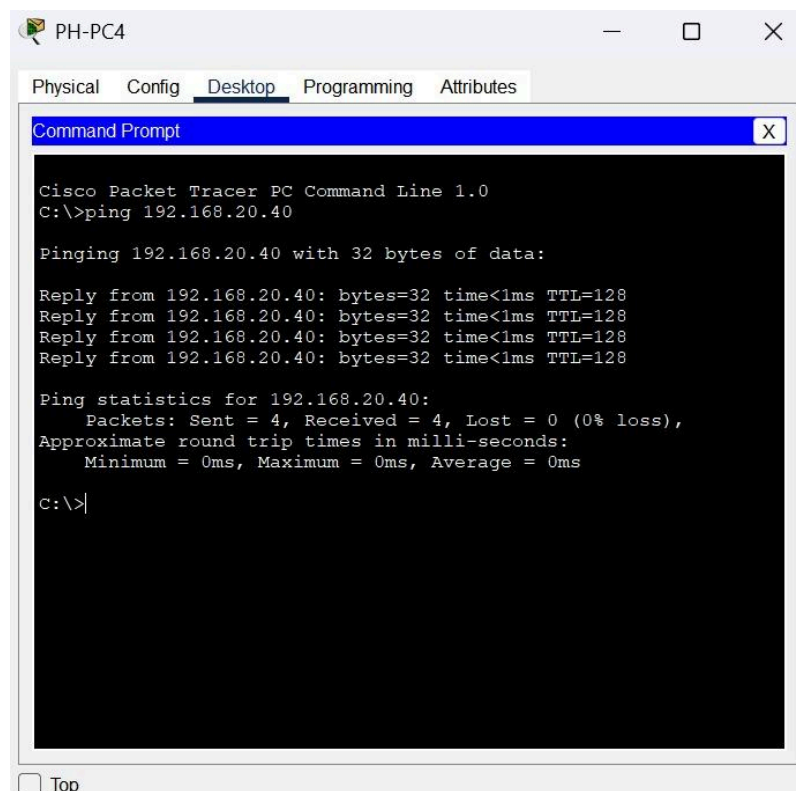
Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time=5ms TTL=128
Reply from 192.168.10.20: bytes=32 time=16ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 5ms

C:\>
```

- Building B



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.40

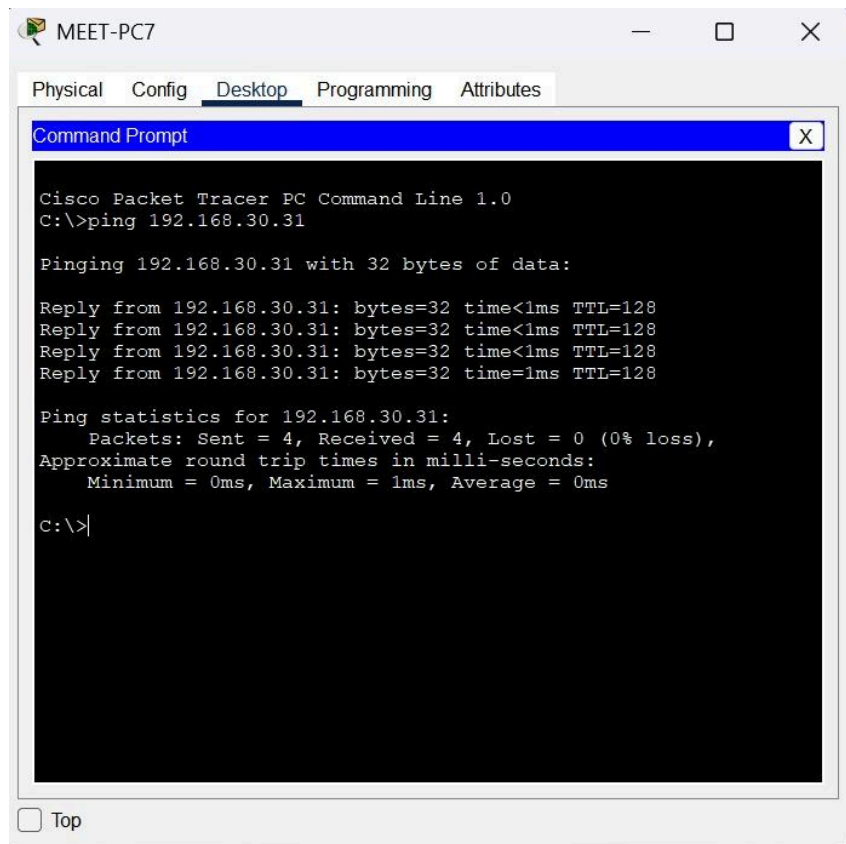
Pinging 192.168.20.40 with 32 bytes of data:

Reply from 192.168.20.40: bytes=32 time<1ms TTL=128
Reply from 192.168.20.40: bytes=32 time<1ms TTL=128
Reply from 192.168.20.40: bytes=32 time<1ms TTL=128
Reply from 192.168.20.40: bytes=32 time<1ms TTL=128

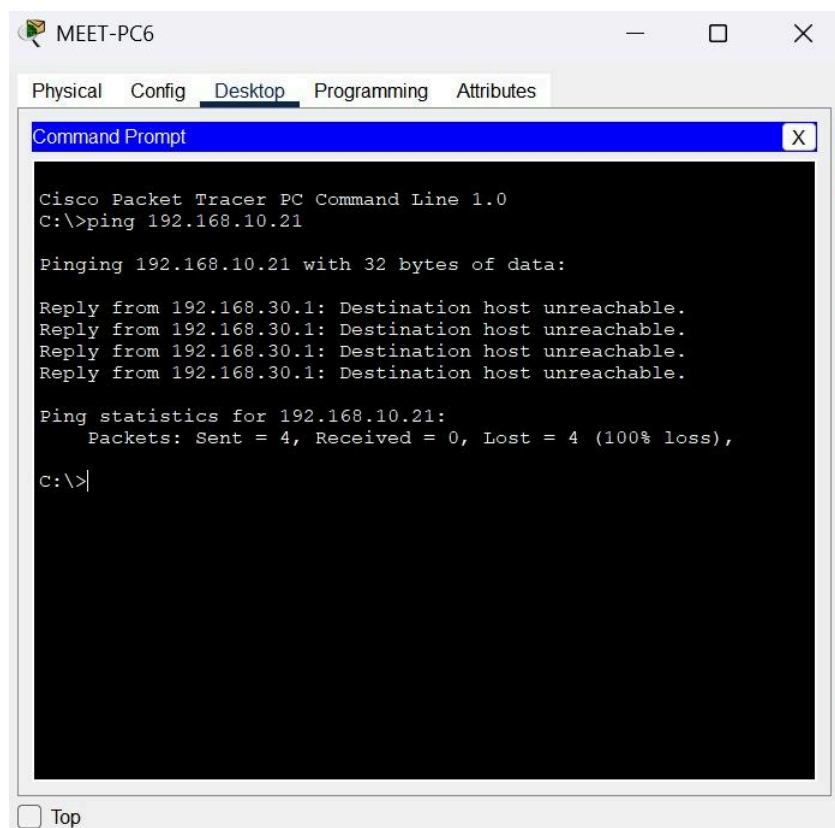
Ping statistics for 192.168.20.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

- Building C



- From building A to C



❖ Scalability for Future Expansion

The design supports future growth because:

- Each building has a /24 subnet (up to 254 hosts)
- Additional switches can be added easily
- Serial links allow more buildings to join the core router
- Modular routers (HWIC-2T) support more WAN interfaces
- Wireless coverage can be expanded with additional APs
- Servers can be upgraded without redesigning the network

❖ Price Estimation

Component	Qty	Estimated Price	Total
2911 Router	4	\$300	\$1200
2960 Switches	6	\$150	\$900
PCs	~80	\$400	\$32,000
Servers	3	\$900	\$2700
Access Points	5	\$100	\$500
Cables & Serial Modules	-	\$200	\$200

Estimated Total Cost ≈ \$37,300

❖ Justification of Component Choices

- Routers & Switches: reliable, scalable, industry-standard, easy to configure
- Serial WAN (HWIC-2T): ideal for building-to-building connectivity
- /24 Subnets: clean logical separation and large capacity
- /30 WAN Links: perfect for point-to-point routing
- APs: provide mobility in waiting rooms, labs, and café
- Servers: support EMR, lab systems, administrative applications
- Dynamic IP (DHCP): reduces manual configuration, easier maintenance

Overall, the chosen components meet the hospital's functional requirements while keeping the design simple, scalable, and easy to maintain.